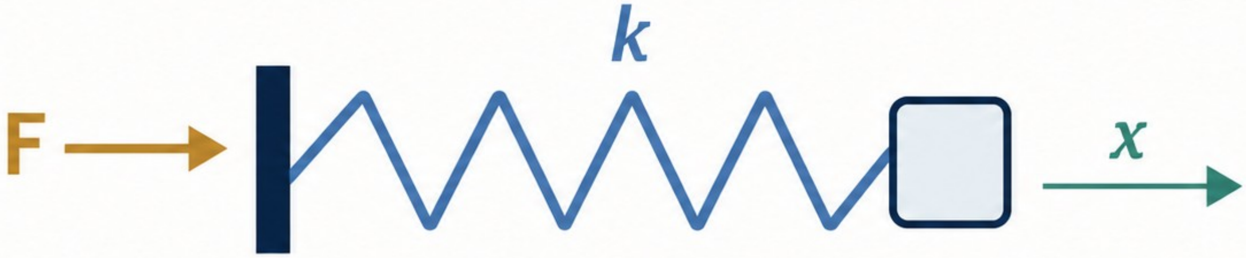
Change one parameter, see the physics

Week 04 | Parameter sweeps and graph interpretation



$$x = F / k$$

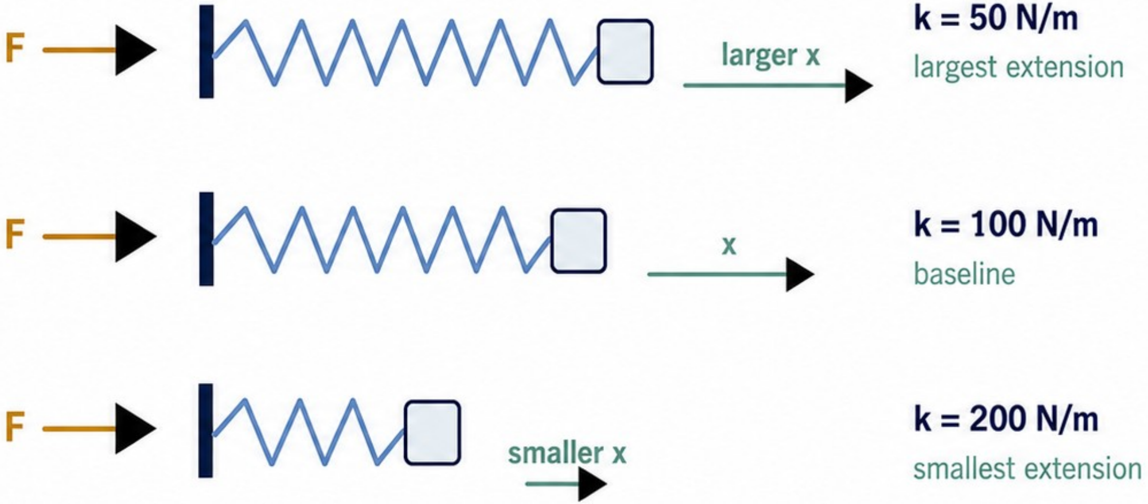
Predict before MATLAB

Model: ideal elastic spring (Hooke's law)

$$F = kx, \qquad x = \frac{F}{k}$$

Same applied force F

compare extension while k changes

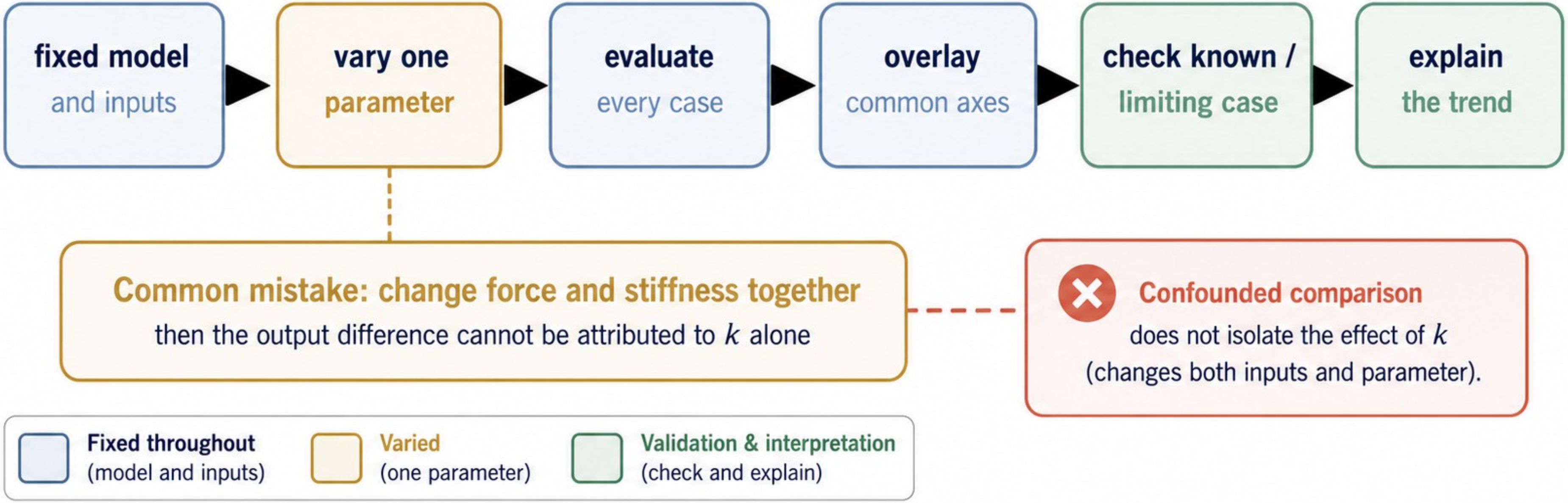


Your predictions

1. Hooke's law gives $x = F/k$.
2. For the same positive force, **smaller k** means **larger extension**.
3. At $F = 10$ N, predict **0.20 m**, **0.10 m**, and **0.05 m** for $k = 50, 100,$ and 200 N/m.
4. At $F = 0$, every ideal-spring case should give $x = 0$.

A sweep is a controlled experiment

- Vary one named parameter: spring stiffness k.
- Hold the force array, model, units, and all other assumptions fixed.
- A comparison that changes force and stiffness together is confounded.
- The fixed-input rule lets the output difference be attributed to k.



Start with the model, variables, and units

- Use an ideal linear spring in its elastic range.
- F $= kx$, so $x = F/k$.
- F_N is the common applied-force array in N.
- k_Npm is the swept stiffness in N/m; extension_m is the output in m.

Physical model: Ideal linear spring

$$F = kx, \qquad x = \frac{F}{k}$$

$F =$ applied force (N)

$k =$ spring stiffness (N/m)

$x =$ spring extension (m)

Variables and roles

Variable	Meaning / Role	Unit
F_N	Applied force (common input for all cases)	N
k_Npm	Spring stiffness (swept parameter)	N/m
$x = F/k$	Spring extension (output for each stiffness case)	m

Baseline_k_Npm = 100 N/m (reference case)

Dimensional check

$$\frac{\mathrm{N}}{\mathrm{N/m}} = \mathrm{m}$$

Force (N) divided by stiffness $(\mathrm{N/m})$ gives extension in metres (m).

Units are consistent.

Plan the computation in plain language

1	**State the model**	Use Hooke's law for an ideal spring: $F = kx$ and $x = F/k$.
2	**Choose swept parameter**	Sweep spring stiffness k $(\mathrm{N/m})$. Define cases: $k = [50 \quad 100 \quad 200]\ \mathrm{N/m}$.
3	**Fix all other inputs**	Force array $F_N = 0\colon 0.5\colon 10\ \mathrm{N}$, equation $x = F/k$, units $(\mathrm{N},\ \mathrm{m})$, assumptions, and any other inputs remain fixed.
4	**Evaluate the same model for every case**	For each k case, compute extension $x = F/k$ at every force value.
5	**Store results in a 3-by-21 arrangement**	One row per stiffness case; one column per force value. Rows correspond to $k = 50,\ 100,\ 200\ \mathrm{N/m}$.
6	**Overlay on common axes with legend**	Plot spring extension x (m) versus applied force F (N) for all cases on the same axes with labels and legend.
7	**Validate a known or limiting case**	Check $F = 0$ gives $x = 0$ for every case. (Zero force → zero extension.)
8	**Explain the trend physically**	Higher stiffness k → smaller extension x for the same force F. (x is inversely proportional to k at fixed F.)

One controlled change rule

This is a controlled computational experiment.

Change (swept)

Spring stiffness

k $(\mathrm{N/m})$

Fixed

- ✓ Force array $F_N = 0\colon 0.5\colon 10\ \mathrm{N}$
- ✓ Model $x = F/k$
- ✓ Units $(\mathrm{N},\ \mathrm{m})$
- ✓ Assumptions
- ✓ Other inputs

Vary one parameter. Hold everything else constant.

Known (limiting) check: $F = 0$ N → $x = 0$ m for all k.
Any curve not passing through the origin is physically incorrect.

Common defect to avoid (runnable but physically wrong):
Using $x = F \times k$ instead of $x = F/k$ gives larger x for larger k.
Trend contradicts Hooke's law and physics.

Store the cases explicitly

```
F_N = 0:0.5:10;
% applied force array (N)

k_Npm = [50 100 200];
% spring stiffness values (N/m)

baseline_k_Npm = 100;
% baseline spring stiffness (N/m)
```

1. **Build one common force array.**
   F_N is the fixed input used for every stiffness case.
2. **Store the stiffness cases explicitly.**
   k_Npm lists the swept values 50, 100, and 200 N/m.
3. **Keep the baseline named so it can be traced later.**
   baseline_k_Npm = 100 N/m.
4. **Units in variable names help keep the physical meaning visible.**
   N/m
   N for Newtons, m for metres.

Evaluate the same model for every case

- Preallocate one row for each stiffness value.
- Each loop pass changes only k_Npm(case_id).
- Use element-wise division for $x = F/k$.
- Row case_id maps to one stiffness case.

MATLAB sweep loop

```
extension_m = zeros(length(k_Npm),length(F_N));
for case_id = 1:length(k_Npm)
    extension_m(case_id,:) = F_N./k_Npm(case_id);
end
```

Result matrix: extension_m (size 3-by-21)

Each row = one stiffness case; each column = one applied force value

	Applied force, F_N (N)						k_Npm(case_id) (N/m)
	0	0.5	1	1.5	⋯	10	
row 1	0	0.01	0.02	0.03	⋯	0.20	50 N/m
row 2	0	0.005	0.01	0.015	⋯	0.10	100 N/m
row 3	0	0.0025	0.005	0.0075	⋯	0.05	200 N/m

21 applied-force values: F_N = 0:0.5:10 N

Known or limiting case check:
At $F = 0$, every row gives $x = 0$.
This confirms the model and computation are consistent.

Interpretation: Larger stiffness (k) produces smaller extension (x) for the same applied force (F), matching Hooke's law $x = F/k$.

Read one row before plotting

- At $F = 10\ \mathrm{N}$, row 1 gives 0.20 m.
- The baseline row gives 0.10 m.
- Row 3 gives 0.05 m.
- The same-input ordering is $x_{50} > x_{100} > x_{200}$.

	k (N/m)	x at $F = 10$ N (m)	case
1	50	**0.20**	lower stiffness
2	100	**0.10**	baseline
3	200	**0.05**	higher stiffness

Same-input comparison at $F = 10\ \mathrm{N}$

All three values are for the same applied force. Read down the column to compare the effect of spring stiffness.

$x_{50} = 0.20$ m (row 1) > $x_{100} = 0.10$ m (baseline) > $x_{200} = 0.05$ m (row 3)

stiffness increases to the right → extension decreases

Overlay every case on common axes

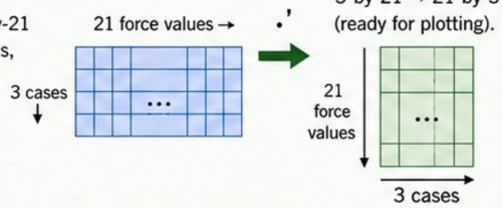
- Use one plot call for all stiffness cases; label the physical axes and identify each stiffness in the legend.
- The transpose changes the stored 3-by-21 arrangement into a plotting arrangement without changing the values.

MATLAB plotting command

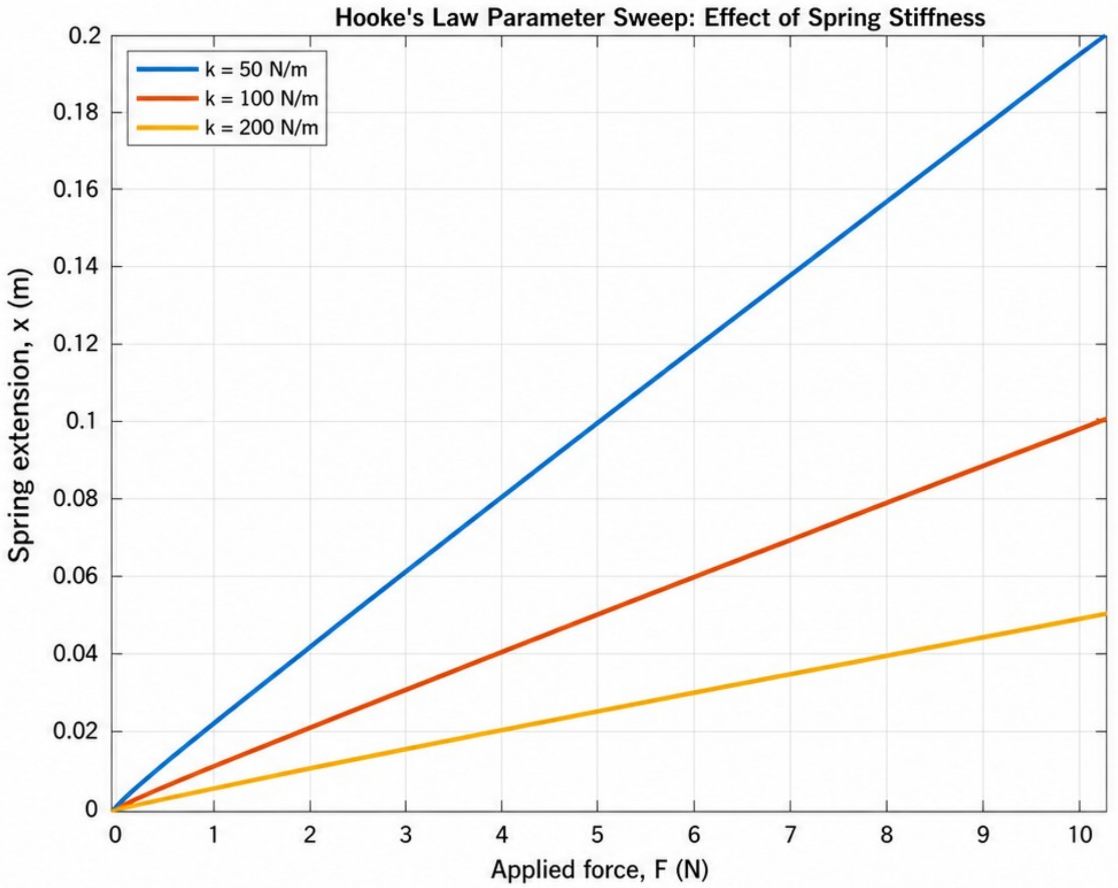
```
plot(F_N,extension_m.','LineWidth',2)
xlabel('Applied force, F (N)')
ylabel('Spring extension, x (m)')
legend('k = 50 N/m','k = 100 N/m', 'k = 200 N/m', ...
     'Location','northwest')
```

Transpose operation (.')

Stored result array: extension_m is 3-by-21 (three stiffness cases, 21 force values).

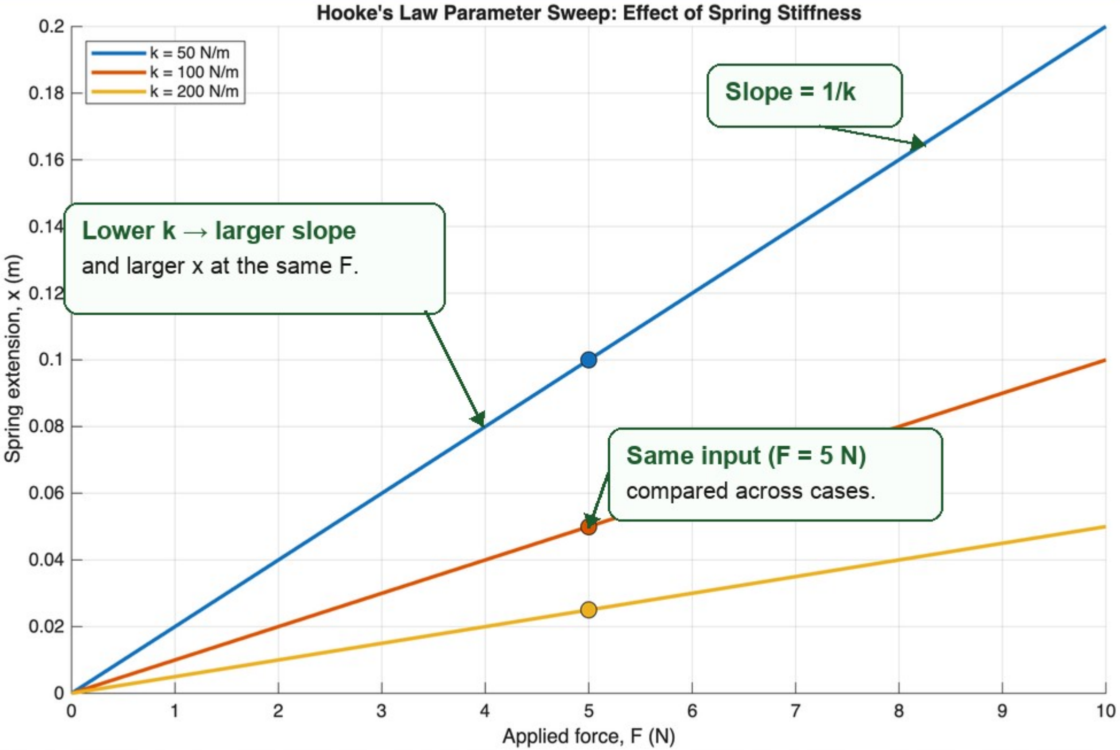


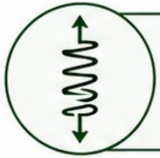
After transpose: 3-by-21 → 21-by-3 (ready for plotting).



The graph is a physics statement

- Each curve is straight because x is proportional to F.
- The slope is $1/k$.
- Smaller k gives a steeper line and larger extension at the same force.
- The graph supports the prediction only because the comparison uses common force inputs.





Physical interpretation: A softer spring (smaller k) stretches more under the same force.
A stiffer spring (larger k) resists more and extends less.

$$x = \frac{F}{k}$$

Validate a known limiting case

- At $F = 0$, Hooke's law requires $x = 0$ for every positive stiffness.
- Read the first result column directly.
- Assert that the largest absolute value is below 1e-12.
- A known limit turns a physical expectation into a reproducible check.

MATLAB validation code (strict)

```
zero_force_extension_m = extension_m(:,1)
assert(max(abs(zero_force_extension_m)) < 1e-12, ...
    'Zero-force limiting-case validation failed.')
```

What this tests: The evaluated model output across all cases.

Not a syntax check: It confirms the physics at a known limit.

Zero-force result (first column of extension_m)

At $F = 0$ N, the spring extension for every stiffness is exactly zero.

$$\text{extension\_m(:,1)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ m}$$

max(abs(extension_m(:,1))) $= 0 <$ 1e-12

Zero-force limiting-case validation passed.

Think: Why must all three values be zero? What would a nonzero value indicate?

A runnable calculation can still be wrong

DIAGNOSIS: MATLAB runs, but the physics is wrong

Wrong vs correct expressions

```
wrong_baseline_extension = F_N.*baseline_k_Npm;
correct_baseline_extension = F_N./baseline_k_Npm;
wrong_final_value = wrong_baseline_extension(end)
correct_final_value_m = correct_baseline_extension(end)
```

Model being used (unchanged)

$$F = kx, \qquad x = \frac{F}{k}$$

Inputs for all cases (fixed)

- Applied force: `F,N = 0:0.5:10 N`
- Stiffness array: `k_Npm = [50 100 200] N/m`
- Baseline stiffness: `baseline_k_Npm = 100 N/m`

PHYSICAL CHECK: The wrong expression fails

Faulty expression: `x = F.*k`

Unit mismatch

F is in N and k is in N/m.

$\mathrm{N} \times (\mathrm{N/m}) = \mathrm{N}^2/\mathrm{m}$,

not metres (m).

Reversed trend

At the same force, larger k gives **larger x** in the wrong model.

Correct model $x = F/k$	Wrong model $x = F.*k$
$k \uparrow \Rightarrow x \downarrow$	$k \uparrow \Rightarrow x \uparrow$
(stiffer → smaller extension)	(stiffer → larger extension)

Repair the expression, then recheck the physics

- At $F = 10$ N, compare correct $x = F/k$ with wrong $x = F.*k$.
- The correct model gives 0.20, 0.10, and 0.05 m; the wrong expression gives 500, 1000, and 2000 $\mathrm{N^2/m}$.
- Repair by tracing the model, units, limiting case, and expected trend.

Comparison at $F = 10$ N

k (N/m)	Correct $x = F/k$ (m)	Wrong $x = F.*k$ ($\mathrm{N^2/m}$)
50	0.20	500
100	0.10	1000
200	0.05	2000

Repair path: from suspicion to correction

1. $F = kx,\ x = \frac{F}{k}$ — **Trace the model**
   Hooke's law: $F = kx,\ x = \dfrac{F}{k}$.
2. $[k] = \mathrm{N/m}$ — **Check units**
   x must be in m.
   The wrong output has units $\mathrm{N^2/m}$.
3. $F = 0 \Rightarrow x = 0$ — **Apply the limiting case**
   At $F = 0$, every case must give $x = 0$.
4. $x \propto \frac{1}{k}$ — **Check the expected trend**
   Larger k should give smaller x.
   The wrong expression gives the opposite.
5. ☑ **Correct the expression**
   Use element-wise division for $x = F/k$.

```
wrong_baseline_extension = F_N.*baseline_k_Npm;
correct_baseline_extension = F_N./baseline_k_Npm;
wrong_final_value = wrong_baseline_extension(end)
correct_final_value_m = correct_baseline_extension(end)
```

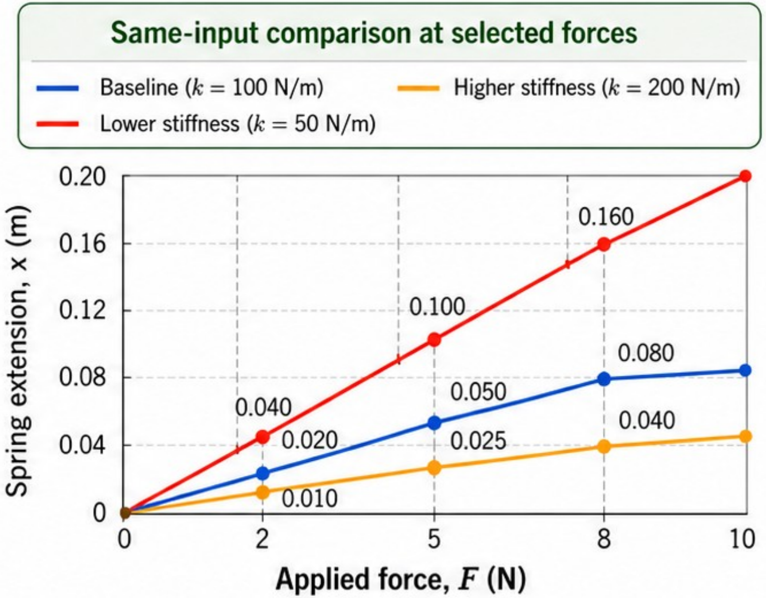
```
correct_baseline_extension = F_N./baseline_k_Npm;
```

Recheck the physics: at $F = 0$, every case gives $x = 0$; at $F = 10$ N, the extensions are 0.20, 0.10, and 0.05 m.

Working exposure: compare selected inputs

- Compare the same selected forces for every stiffness case.
- At $F = [2\ \ 5\ \ 8]$ **N**, the values follow the same inverse-stiffness ordering.
- A table and a graph are two views of the same controlled experiment.

F (N)	$k = 50$ N/m	$k = 100$ N/m	$k = 200$ N/m
2	0.040	0.020	0.010
5	0.100	0.050	0.025
8	0.160	0.080	0.040



At the same F, extensions follow the extension order: $x_{50} > x_{100} > x_{200}$.

Exit ticket — explain a fair sweep to a future you

PREDICT
the trend from the equation.

Hooke's law

$$F = kx \quad \text{and} \quad x = \frac{F}{k}$$

For a fixed $F > 0$:

$$k\uparrow \quad \longrightarrow \quad x\downarrow$$

Stiffer spring → smaller extension.

CONTROL
one changed parameter and the fixed inputs.

Swept parameter:

spring stiffness, k (N/m)

$$k_{Npm} = [50\ \ 100\ \ 200]\ \text{N/m}$$

Fixed model and input array:

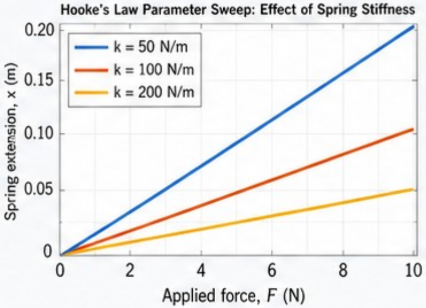
Ideal Hooke's law, $x = F/k$

$$F_N = 0{:}0.5{:}10\ \text{N}$$

Same model, same F, different k.

PLOT
labelled cases on common axes.

Common axes:



Compare cases at the same F.

CHECK & INTERPRET
a known or limiting case, then the physical result.

Known or limiting case:

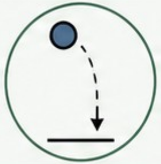
At $F = 0$ N, all cases give $x = 0$ m. (Passes.)

Interpretation:

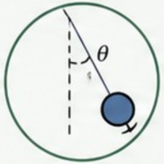
For the same force, a larger k produces a smaller extension. Matches $x = F/k$.

Transfer the workflow
to analogous supplied models.

Ohmic resistor
$V = IR$



vertical motion
$s = ut + \frac{1}{2}at^2$



small-angle pendulum
$T = 2\pi\sqrt{L/g}$